

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R, S) / S1 (PT) (S, FE) Examination December 2023 (2019 Scheme)

Course Code: EET205**Course Name: ANALOG ELECTRONICS**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 List the factors to be considered while selecting the operating point of a transistor in an amplifier circuit. (3)
- 2 What is the role of emitter bypass capacitor and coupling capacitors in an amplifier circuit? (3)
- 3 Sketch the transfer characteristics of an n channel JFET given $I_{DSS} = 15\text{mA}$ and $V_p = -6\text{V}$. (3)
- 4 Define transconductance of JFET. (3)
- 5 State and explain Barkhausen's criteria for sustained oscillations. (3)
- 6 A feedback amplifier has a voltage gain of 300 without feedback. Determine the voltage gain with feedback if feedback ratio is 0.2. (3)
- 7 Write any six characteristics of an ideal operational amplifier. (3)
- 8 Design a non-inverting amplifier for a gain of 12. (3)
- 9 Derive the output voltage equation of an ideal differentiator with circuit diagram. (3)
- 10 Explain zero crossing detector circuit with diagram and waveforms. (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Design a voltage divider bias circuit for an operating point of $I_C = 2\text{mA}$ and $V_{CE} = 8\text{V}$. Given $V_{CC} = 20\text{V}$, $\beta = 100$, $V_{BE} = 0.7\text{V}$, $V_E = 2\text{V}$. (10)
- (b) Define stability factor. Derive an expression for stability factor. (4)
- 12 (a) Draw the small signal ac equivalent circuit of common emitter amplifier using h parameters. Also derive the expressions for voltage gain, current gain, input impedance and output impedance. (10)
- (b) How bias compensation is done using diodes? (4)

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Module 2

- 13 (a) Explain, with neat diagram, the construction and working of n-channel enhancement mode MOSFET. (10)
- (b) Explain the hybrid pi model of BJT. (4)
- 14 (a) Draw the frequency response of common emitter amplifier. State the reasons for reduction of gain at low frequencies and high frequencies. (7)
- (b) Using small signal equivalent circuit, derive the expression for voltage gain of common drain JFET amplifier. (7)

Module 3

- 15 (a) Define conversion efficiency of power amplifier. Derive the maximum conversion efficiency of transformer coupled class A power amplifier. (7)
- (b) Design a RC phase shift oscillator circuit for a frequency of 2 kHz. (7)
- 16 (a) Derive the expression for frequency of oscillation of Wein bridge oscillator circuit with neat diagram. (10)
- (b) Mention the merits and demerits of transformer coupled amplifier. (4)

Module 4

- 17 (a) Determine the output voltage of an op-amp for input voltages of $V_1 = 120 \text{ mV}$ and $V_2 = 110 \text{ mV}$. The amplifier has a differential gain of $A_d = 4000$ and the value of CMRR is 100. (7)
- (b) Explain the concept of virtual ground. What is the limiting value of output voltage of an op-amp. Justify your answer. (7)
- 18 (a) Design a summing amplifier that satisfies the output equation $V_0 = -(V_1 + 5V_2 + 2V_3)$. (7)
- (b) Explain instrumentation amplifier with diagram. (7)

Module 5

- 19 (a) Explain the working of Schmitt trigger circuit with diagram and waveforms. (7)
- (b) Explain the working of triangular wave generator using op-amp with diagram and waveforms. (7)
- 20 (a) Draw and explain the internal diagram of 555 timer IC. (7)
- (b) Design an astable multivibrator using 555 timer for an output wave of 60% duty cycle at a frequency of 2 kHz. (7)